



CHAPTER cxlii.

An Act to provide in the case of certain gas companies for the adoption of the Metropolitan Argand Burner No. 2 as a standard burner in substitution for the various burners now in use for the official testing of the illuminating power of gas supplied by them and for other purposes. A.D. 1910.
[28th November 1910.]

WHEREAS by the City of London Gas Act 1868 it was enacted that the burner to be used for testing the illuminating power of gas supplied in the city of London should be “such as shall be the most suitable for obtaining from the gas the greatest amount of light and be practicable for use by the consumer”:

And whereas the Gasworks Clauses Act 1871 enacts that the illuminating power of gas supplied by the various gas undertakings of the kingdom should be tested by burners to be prescribed by the special Act relating to each company none of which burners (with the exception of the burner prescribed in Gas Acts of recent date and hereinafter referred to) fulfil the conditions prescribed by Parliament as aforesaid or meet the requirements essential for a fair and scientifically accurate test for illuminating power:

And whereas the metropolitan gas referees after careful experiment have satisfied themselves that the burner known as the Metropolitan Argand Burner No. 2 is the best known burner for the testing of illuminating power of gas and they have since the year 1906 adopted the same for the purpose of testing in accordance with the conditions and in the manner hereinafter set forth the illuminating power of gas supplied by the three metropolitan gas companies:

And whereas it has been proved that the said burner gives a fair and true result for all qualities of gas up to an illuminating

A.D. 1910. power of twenty candles which limit covers the statutory obligations of all the gas companies enumerated in the First Schedule to this Act (hereinafter called "the Promoting Companies") :

And whereas the said burner is now invariably prescribed in all Acts and Provisional Orders conferring powers of supply upon gas companies and local authorities as also in Acts and Provisional Orders promoted by existing companies and local authorities for further powers :

And whereas the Promoting Companies are desirous of having the said burner substituted for the burner prescribed in the Acts or Orders relating to their respective undertakings and it is expedient to enact that the said burner should be substituted accordingly :

And whereas the objects aforesaid cannot be attained without the authority of Parliament :

May it therefore please Your Majesty that it may be enacted and be it enacted by the King's most Excellent Majesty by and with the advice and consent of the Lords Spiritual and Temporal and Commons in this present Parliament assembled and by the authority of the same as follows (that is to say) :—

Short title.

1. This Act may be cited for all purposes as the Gas Companies (Standard Burner) (No. 2) Act 1910.

Interpretation.

2. "The prescribed burner" means the burner prescribed pursuant to the provisions of the Gasworks Clauses Act 1871 in any Act or Order relating respectively to the Promoting Companies.

"The standard burner" means the Metropolitan Argand Burner No. 2 described by the metropolitan gas referees in their notification for the year 1908 (a model of which burner has been deposited with the warden of the standards) or any other burner which the Board of Trade may approve under the provisions of this Act as being equally or more suitable for the testing for illuminating power.

Commencement of Act.

3. This Act shall come into operation on the first day of January one thousand nine hundred and eleven.

Substitution of standard burner for testing gas supplied by Promoting Companies.

4. Notwithstanding anything in any Acts or Orders relating to the Promoting Companies contained the testing for illuminating power of the gas supplied by the Promoting Companies shall subject to the provisions of this Act be by means of the standard burner which shall be deemed to be the prescribed

A.D. 1910.

burner within the meaning of the Gasworks Clauses Act 1871 for the purposes of any test for illuminating power under that Act and the Acts and Orders set forth in the second column of the First Schedule to this Act and such Acts and Orders are hereby severally amended and shall be read and have effect accordingly Provided that the Board of Trade may on the application of the company or the local authority concerned approve the use of any other burner the chimney and burner holes of which may have other dimensions and which may appear to the Board to be equally or more suitable for the testing Provided further that such other burner shall be the most suitable for obtaining from the gas the greatest amount of light and be practicable for use by the consumer.

5. The following conditions shall be observed when testing with the standard burner:—

Defining
conditions
and method
of testing
with stand-
ard burner.

- (1) The glass chimney to be used with the standard burner shall be six inches in length by one inch and five-eighths of an inch in internal diameter:
- (2) The gas shall be consumed in the standard burner at the rate of five cubic feet per hour and the air supply thereto shall be adjusted according to the regulations contained in Part I. of the Second Schedule to this Act:
- (3) If the gas is so rich that it cannot be burned with the full air supply and at the prescribed rate of consumption without tailing above the chimney the rate of consumption shall be reduced until the flame burns properly within the chimney as described in Part I. of the said Second Schedule The rate of consumption shall then be carefully ascertained and from this the illuminating power of the gas under test shall be calculated by simple proportion for any reduction of such rate of consumption below five cubic feet per hour:
- (4) Proper correction shall be made according to the method set forth in Parts II. and III. of the Second Schedule to this Act for variation from standard atmospheric conditions of thirty inches height of mercury pressure and sixty degrees Fahrenheit.

6. All costs charges and expenses of and incident to the preparing for obtaining and passing of this Act or otherwise in relation thereto shall be paid by the Promoting Companies.

Costs of Act.

A.D. 1910. The SCHEDULES referred to in the foregoing Act.

FIRST SCHEDULE.

Name of Company.	Act or Order.
Berkhampstead Gas Company - -	Berkhampstead Gas Act 1905.
Bournemouth Gas and Water Company -	Bournemouth Gas and Water Act 1878.
British Gas Light Company Limited (in respect of their Staffordshire Potteries Trowbridge and Holywell Stations).	British Gaslight Company Limited (Staffordshire Potteries) Act 1880. Trowbridge Gas Order 1880. Holywell Gas Order 1880.
Cambridge University and Town Gas Light Company.	Cambridge University and Town Gas Act 1899.
Chigwell Loughton and Woodford Gas Company.	Chigwell Loughton and Woodford Gas Act 1873.
Faversham Gas Company - -	Faversham Gas Act 1889.
Harrow and Stanmore Gas Company -	Harrow and Stanmore Gas Act 1894.
Hatfield Gas Company Limited - -	Hatfield Gas Order 1888.
Hemel Hempsted District Gas Company	Hemel Hempsted District Gas Act 1878.
Herne Bay Gas and Coke Company Limited.	Herne Bay Gas Order 1899.
Newmarket Gas Company - -	Newmarket Gas Company's Act 1895.
Ormskirk Gas Light Company - -	Ormskirk Gas Light Act 1892.
Prescot Gas Company - -	Prescot Gas Act 1880.
Radcliffe and Pilkington Gas Company -	Radcliffe and Pilkington Gas Act 1869. Radcliffe and Pilkington Gas Act 1878.
Reading Gas Company - -	Reading Gas Act 1880.
Shrewsbury Gas Light Company -	Shrewsbury Gas Act 1878.
Tunbridge Wells Gas Company	Tunbridge Wells Gas Act 1887.
Waltham Abbey and Cheshunt Gas and Coke Company.	Waltham Abbey and Cheshunt Gas Order 1881.
West Kent Gas Company - -	West Kent Gas Act 1867.
Worthing Gas Light and Coke Company	Worthing Gas Act 1875.

SECOND SCHEDULE.

PART I.

REGULATIONS FOR USING THE STANDARD BURNER.

The gas to be examined must be kept alight in the burner at or about the rate of five cubic feet per hour and with the air regulator screwed down so as to be fully open for at least thirty minutes before any testing is commenced.

When the examination is about to be made and the gas consumption has been adjusted to five cubic feet per hour the air regulator of the burner must be screwed upwards until the flame rises in the chimney as

high as possible without smoking and no more air is to be admitted by the regulator than is necessary to accomplish this The photometrical readings for illuminating power are then to be made in the usual manner. A.D. 1910.

From time to time the accuracy of the burner holes must be tested by two gauges having respective diameters of 0·058 inch and 0·062 inch The holes must be of such size that the smaller only of the two gauges can be inserted therein.

The bottom edge of the glass chimney is to be ground true so that it stands upright upon the burner and a clean one must be used for each testing.

PART II.

REGULATIONS AS TO CORRECTING TO STANDARD TEMPERATURE AND PRESSURE.

At the time of each testing the gas examiner shall observe and record the temperature of the gas as shown by the thermometer attached to the meter and also the height of the barometer The volume of the gas operated upon during the testing shall be corrected from these data to the standard of comparison namely for the barometer 30 inches of mercury and for the thermometer 60 degrees Fahrenheit The correction may be made by means of the Table of Numbers given in Part III. of this schedule If for example the thermometer reading is 54 degrees and the barometer reading 30·3 inches the tabular number will be 1·025 and multiplying by this figure the observed quantity of gas consumed per hour will give the corrected volume of gas so consumed that is to say the volume it would have occupied when measured over water at the standard temperature and pressure.

The originally observed illuminating power shall then be corrected by simple proportion for any variation between the observed rate of consumption and the true rate of consumption as ascertained by such correction to standard temperature and pressure.

PART III.

TABLE TO FACILITATE THE CORRECTION OF THE VOLUME OF GAS AT DIFFERENT TEMPERATURES AND UNDER DIFFERENT ATMOSPHERIC PRESSURES.

The numbers in the following table have been calculated from the formula $n = \frac{17 \cdot 64 (h-a)}{460 + t}$ where h is the height of the barometer in inches t the temperature on the Fahrenheit scale and a the tension of aqueous vapour at t degrees If v is any volume at t degrees and h inches pressure and V the corresponding volume at 60 degrees and 30 inches pressure $V = vn$.

BAR.	THERMOMETER			— FAHRENHEIT.				54°	56°	58°	60°
	40°	42°	44°	46°	48°	50°	52°				
28·0	·979	·974	·970	·965	·960	·956	·951	·946	·942	·937	·932
28·1	·983	·978	·973	·969	·964	·959	·955	·951	·945	·941	·936
28·2	·986	·981	·977	·972	·967	·963	·958	·953	·949	·944	·939
28·3	·990	·985	·980	·976	·971	·966	·961	·957	·952	·947	·942
28·4	·993	·988	·984	·979	·974	·970	·965	·960	·955	·951	·946
28·5	·997	·992	·987	·983	·978	·973	·968	·964	·959	·954	·949
28·6	1·001	·995	·991	·986	·981	·977	·972	·967	·962	·958	·953
28·7	1·004	·999	·994	·990	·985	·980	·975	·970	·966	·961	·956
28·8	1·007	1·003	·998	·993	·988	·984	·979	·974	·969	·964	·959
28·9	1·011	1·006	1·001	·997	·992	·987	·982	·977	·973	·968	·963
29·0	1·014	1·010	1·005	1·000	·995	·990	·986	·981	·976	·971	·966
29·1	1·018	1·013	1·008	1·004	·999	·994	·989	·984	·979	·975	·969
29·2	1·021	1·017	1·012	1·007	1·002	·997	·992	·988	·982	·978	·973
29·3	1·025	1·020	1·015	1·011	1·006	1·001	·996	·991	·986	·981	·976
29·4	1·028	1·024	1·019	1·014	1·009	1·004	·999	·995	·990	·985	·980
29·5	1·032	1·027	1·022	1·018	1·013	1·008	1·003	·998	·993	·988	·983
29·6	1·036	1·031	1·026	1·021	1·016	1·011	1·006	1·001	·996	·992	·986
29·7	1·039	1·034	1·029	1·025	1·019	1·015	1·010	1·005	1·000	·995	·990
29·8	1·043	1·038	1·033	1·028	1·023	1·018	1·013	1·008	1·003	·998	·993
29·9	1·046	1·041	1·036	1·031	1·026	1·022	1·017	1·012	1·007	1·002	·997
30·0	1·050	1·045	1·040	1·035	1·030	1·025	1·020	1·015	1·010	1·005	1·000
30·1	1·053	1·048	1·043	1·038	1·033	1·029	1·024	1·019	1·014	1·009	1·003
30·2	1·057	1·052	1·047	1·042	1·037	1·032	1·027	1·022	1·017	1·012	1·007
30·3	1·060	1·055	1·050	1·045	1·040	1·036	1·030	1·025	1·020	1·015	1·010
30·4	1·064	1·059	1·054	1·049	1·044	1·039	1·034	1·029	1·024	1·019	1·014
30·5	1·067	1·062	1·057	1·052	1·047	1·042	1·037	1·032	1·027	1·022	1·017
30·6	1·071	1·066	1·061	1·056	1·051	1·046	1·041	1·036	1·031	1·026	1·020
30·7	1·074	1·069	1·064	1·059	1·054	1·049	1·044	1·039	1·034	1·029	1·024
30·8	1·078	1·073	1·068	1·063	1·058	1·053	1·048	1·043	1·037	1·032	1·027
30·9	1·081	1·076	1·071	1·066	1·061	1·056	1·051	1·046	1·041	1·036	1·031
31·0	1·085	1·080	1·075	1·070	1·065	1·060	1·055	1·049	1·044	1·039	1·034

BAR.	Ther. 62°	64°	66°	68°	70°	72°	74°	76°	78°	80°	82°	84°
28·0	·927	·922	·917	·912	·907	·902	·897	·892	·887	·881	·875	·870
28·1	·930	·926	·921	·916	·911	·905	·900	·895	·890	·884	·879	·873
28·2	·934	·929	·924	·919	·914	·909	·904	·898	·893	·887	·882	·876
28·3	·937	·932	·928	·922	·917	·912	·907	·902	·896	·891	·885	·880
28·4	·941	·936	·931	·926	·921	·915	·910	·905	·900	·894	·888	·883
28·5	·944	·939	·934	·929	·924	·919	·914	·908	·903	·897	·892	·886
28·6	·947	·943	·938	·932	·927	·922	·917	·912	·906	·901	·895	·889
28·7	·951	·946	·941	·936	·931	·925	·920	·915	·909	·904	·898	·893
28·8	·954	·949	·944	·939	·934	·929	·924	·918	·913	·907	·901	·896
28·9	·958	·953	·948	·942	·937	·932	·927	·921	·916	·910	·905	·899
29·0	·961	·956	·951	·946	·941	·935	·930	·925	·919	·914	·908	·903
29·1	·964	·959	·954	·949	·944	·939	·933	·928	·923	·917	·911	·906
29·2	·968	·963	·958	·952	·947	·942	·937	·931	·926	·920	·914	·909
29·3	·971	·966	·961	·956	·950	·945	·940	·935	·929	·923	·918	·912
29·4	·975	·969	·964	·959	·954	·949	·943	·938	·932	·927	·921	·915
29·5	·978	·973	·968	·962	·957	·952	·947	·941	·936	·930	·924	·919
29·6	·981	·976	·971	·966	·960	·955	·950	·944	·939	·933	·927	·922
29·7	·985	·980	·974	·969	·964	·959	·953	·948	·942	·937	·931	·925
29·8	·988	·983	·978	·972	·967	·962	·957	·951	·946	·940	·934	·928
29·9	·991	·986	·981	·976	·970	·965	·960	·954	·949	·943	·937	·932
30·0	·995	·990	·985	·979	·974	·968	·963	·958	·952	·946	·941	·935
30·1	·998	·993	·988	·983	·977	·972	·966	·961	·955	·950	·944	·938
30·2	1·002	·996	·991	·986	·980	·975	·970	·964	·959	·953	·947	·941
30·3	1·005	1·000	·995	·989	·984	·978	·973	·968	·962	·956	·950	·945
30·4	1·008	1·003	·998	·993	·987	·982	·976	·971	·965	·959	·954	·948
30·5	1·012	1·006	1·001	·996	·990	·985	·980	·974	·969	·963	·957	·951
30·6	1·015	1·010	1·005	·999	·994	·988	·983	·977	·972	·966	·960	·954
30·7	1·018	1·013	1·008	1·003	·997	·992	·986	·981	·975	·969	·963	·957
30·8	1·022	1·017	1·011	1·006	1·000	·995	·990	·984	·978	·972	·967	·961
30·9	1·025	1·020	1·015	1·009	1·004	·998	·993	·987	·982	·976	·970	·964
31·0	1·029	1·023	1·018	1·013	1·007	1·002	·996	·991	·985	·979	·973	·967

